

SHAMUTHRIKA SP

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Portfolio : <https://portfolio-914r.vercel.app/>

PROFESSIONAL SUMMARY

Full-Stack Engineer with specialized expertise in multi-device automation frameworks, POS terminal test infrastructure, and scalable web architectures. Proven ability to ship end-to-end applications spanning custom network routing pipelines, interactive UI dashboards, and production-ready full-stack systems built with React, Node.js, and machine learning APIs. Four hackathon Finalist designations under competitive time constraints.

EDUCATION

B.E Computer Science Engineering

Chennai Institute of Technology, Chennai, Tamil Nadu

CGPA: 8.82

Batch:2024-2028

PROFESSIONAL EXPERIENCE

- **Data Analyst & QA Testing Intern, Dejavoo** (Onsite) Apr 22 - Jul 12
Architected a multi-device Python and uiautomator2 engine to run bulk and solo transaction regression loops across L200, P3, and wireless Q2 terminal architectures. Conducted deep-dive technical root-cause analyses on application canvas obfuscations to define framework blueprints for hybrid visual-recognition strategies.
- **Data Science Intern, GyanData** (Onsite, 4 weeks) Nov 3-28, 2025
Mined and sanitized data from annual reports for the Annual Report Digitalization Project, organizing complex financial information into a consolidated main database to enable scalable analysis and reporting. Integrated NLP capabilities with SQL querying to automate data extraction and processing, addressing real-world challenges in regulatory compliance and document digitization for enterprise clients.
- **Software Developer Intern, Fourth Dimension Software Systems** (Onsite) Jun 5-25, 2025
Developed core features for an Inventory Management System, including data modeling, CRUD operations, and UI integration to streamline stock tracking and operations. Delivered a fully functional basic working model as the internship capstone, demonstrating independent problem-solving and end-to-end software development proficiency.

PROJECTS

1. *SilentVoice* :

- a real-time AI communication tool for the deaf-mute community, converting **Indian Sign Language (ISL) to speech/text**. Engineered high-performance gesture recognition using **TensorFlow.js** to enable seamless, low-latency in-browser translation.
- Designed a **bi-directional system** that translates voice/text into sign animations, ensuring affordable and accessible education for students.

2. *A.R.C*:

- A robust management platform using **React.js and Tailwind CSS** to streamline the tracking and administration of student records for department heads. Engineered **advanced querying and multi-tier filtering logic** to allow for real-time data retrieval and manipulation, improving administrative efficiency by providing instant access to student metrics.
- Architected a secure backend and database integration with **Node.js & Firebase** to manage complex student datasets, ensuring data integrity while deploying the solution via Vercel for high availability.

3. *Chemick*:

- An AI-powered scanning application using **React Native and Gemini Vision API** to instantly decode and describe chemical compositions in medicines and consumer products. Engineered a multi-modal analysis engine that identifies active ingredients via **OCR** and provides automated safety guidelines, usage instructions, and potential contraindications for healthcare and household items. Optimized the image-processing pipeline using **TensorFlow.js**.

TECHNICAL SKILLS

- **Languages:** Python (3.10), C++ (14.2), Java, JavaScript, TypeScript, SQL
- **Frameworks, Libraries & Web Engines:** React.js (19), Next.js, Node.js, Express.js, React Native, Streamlit, Tailwind CSS
- **Databases & Cloud backends:** MySQL, MongoDB, Firebase, Node.js Backends
- **Machine Learning & AI:** TensorFlow.js, NLP, OCR, Image-Processing Pipelines, NumPy, Matplotlib

CERTIFICATIONS

- **Google** - Data Analytics Certification.
- **AICTE** - IOT Cloud Engineer, Generative AI Virtual Internship.
- **NPTEL** - Developing Soft Skills and Personality, Introduction to Internet of things, Data Analytics with Python
- **MongoDB** - AI fundamentals

ACHIEVEMENTS

- **ICPC AlgoQueen 2026 - State Champion (2nd in Tamil Nadu) and 59th in National ranking.**
- **Leetcode Knight Badge:** 790+ problems solved with max rating 2000+ in contests.
- **Hackathons finalists:** Visa AI 242025, IMC prosperity 2026, Hack the Flame 2025.